

1. Administrative & Study Identification

Full Study Title: Two Part (Double-blind Inclisiran Versus Placebo [Year 1] Followed by Open-label Inclisiran [Year 2]) Randomized Multicenter Study to Evaluate Safety, Tolerability, and Efficacy of Inclisiran in Children (2 to Less Than 12 Years) With Homozygous Familial Hypercholesterolemia and Elevated LDL-cholesterol **Short Title/Acronym:** Inclisiran in Children With HoFH **Protocol Number:** 1103090460876593113 **NCT Number:** NCT06597006 **Sponsor Name:** Novartis **Investigational Product:** Inclisiran (Trade Name: Leqvio, ATC Code: C10AX16) **Phase of Development:** Phase III **Trial Status:** Recruiting **Medical Monitor:** Novartis Clinical Medical Director

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the effect of inclisiran compared to placebo on reducing LDL-C (percentage change from baseline) at Day 330 (Year 1). **Secondary Objectives:** To evaluate time-adjusted percent change in LDL-C from baseline up to Day 330; and to assess the percent change in LDL-C, total cholesterol, non-HDL-C, triglycerides, HDL-C, VLDL-C, and PCSK9 from baseline to each assessment time up to Day 720 (Year 2).

2.2 Background & Rationale To evaluate the safety, tolerability, and efficacy of inclisiran in a vulnerable pediatric population (children aged 2 to <12 years) with homozygous familial hypercholesterolemia (HoFH) and elevated LDL-C despite stable standard of care background lipid-lowering therapy. Managing HoFH early is a critical medical necessity to prevent premature cardiovascular events.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (during Year 1), followed by an Open-label active treatment phase (Year 2) **Blinding:** Double-blind (Year 1), transitioning to Open-label (Year 2) **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: 9 evaluable patients **Number of Sites:** Multicenter (Global) **Estimated Study Start:** Q4 2024 **Estimated Primary Completion:** Q4 2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female participants, 2 to <12 years of age at screening. 2. HoFH diagnosed by genetic confirmation (excluding known null/negative mutations in both LDLR alleles). 3. Fasting LDL-C >130 mg/dL (3.4 mmol/L) at screening. 4. On an optimal dose of statin (investigator's discretion), unless statin intolerant, with or without other lipid-lowering therapy (e.g., ezetimibe). Participants on lipid-lowering therapies must be on a stable dose for ≥ 30 days before screening with no planned medication or dose changes. 5. Participants on a documented regimen of LDL-apheresis for ≥ 3 months before screening are allowed to continue; the apheresis schedule/settings/duration must be stable prior to screening, must not change during the double-blind period, and must permit that an apheresis coincides with each study visit.

4.2 Exclusion Criteria 1. Documented evidence of a null (negative) mutation in both LDLR alleles. 2. Previous treatment (within 90 days of screening) with monoclonal antibodies directed towards PCSK9 or history of poor response to them (e.g., <15% reduction in LDL-C). 3. Treatment with mipomersen or lomitapide within 5 months of screening. 4. Secondary hypercholesterolemia (e.g., hypothyroidism or nephrotic syndrome), HeFH, or body weight <16 kg for participants 6 to <12 years or <11 kg for participants 2 to <6 years at screening. 5. Active liver disease defined as any known current infectious, neoplastic, or metabolic pathology of the liver or unexplained ALT/AST elevation >3x ULN, or total bilirubin elevation >2x ULN (except patients with Gilbert's syndrome). 6. Pregnant or nursing females. 7. Recent and/or planned use of other investigational medicinal products or devices.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Dosed per protocol body-weight tiers; Part 1 (Year 1) utilizes Inclisiran vs. Placebo; Part 2 (Year 2) utilizes Open-label Inclisiran. **Route of Administration:** Subcutaneous Injection **Duration of Treatment:** 720 Days (approx. 2 years)

5.2 Concomitant & Prohibited Therapy Permitted: Stable standard of care background lipid-lowering therapies (e.g., statins, ezetimibe) and established stable LDL-apheresis. **Prohibited:** PCSK9 monoclonal antibodies (within 90 days prior), mipomersen, and lomitapide (within 5 months prior). Medication dose changes during the study are restricted.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	
Screening	Day -30 to 0	Informed Consent/ Assent, Genetic confirmation, Fasting LDL-C Labs, Weight check	
Baseline	Day 0	Randomization, First Dose (Inclisiran/ Placebo), Lipid Panel	
Interim 1	Day 90 ± 7	Safety Labs, Efficacy Assessment (LDL-C monitoring)	
End of Year 1	Day 330	Primary Endpoint Assessment (LDL-C % change), Transition to Open-label	
End of Study	Day 720	Final Vital Signs, Full Lipid Panel, PCSK9 assessment, IP Accountability	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Percentage change in LDL-C from baseline to Day 330 (Year 1). **Measurement Method:** Central Laboratory Analysis of fasting lipid panel.

7.2 Secondary & Exploratory Endpoints 1. Time-adjusted percent change in LDL-C from baseline after Day 90 and up to Day 330 (Year 1). 2. Percent change in LDL-C, total cholesterol, non-HDL-C, triglycerides, HDL-C, and VLDL-C from baseline to each assessment time up to Day 720 (Year 2). 3. Percent change in PCSK9 from baseline to each assessment time up to Day 720 (Year 2).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard clinical evaluation, physical exams, and lab monitoring across scheduled visits.

Serious Adverse Events (SAEs): Reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** External independent Data Monitoring Committee (iDMC) to review unblinded safety and efficacy data due to the vulnerable pediatric population.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for all adverse event summaries. **Sample Size**

Justification: N=9 target. Rare disease constraint (pediatric HoFH); powered based on expected large effect sizes of inclisiran on LDL-C versus placebo. **Handling of Missing Data:** Mixed-effect Model Repeated Measure (MMRM) or Multiple Imputation for missing continuous variables across the 720-day span.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits by the Sponsor (Novartis) to ensure protocol adherence and pediatric safety. **Audit Trail:** Electronic Data Capture (EDC) system with full audit trail, eCRF locking, and automated data clarification forms.

11. Study Identifiers & Governance

Primary ID: 1103090460876593113 **Registry Number:** NCT06597006

Sponsor/Lead Organization: Novartis **Study Title:** Study to Evaluate Safety, Tolerability and Efficacy of Inclisiran in Children With Homozygous Familial Hypercholesterolemia **Official Regulatory Title:** Two Part (Double-blind Inclisiran Versus Placebo [Year 1] Followed by Open-label Inclisiran [Year 2]) Randomized Multicenter Study to Evaluate Safety, Tolerability, and Efficacy of Inclisiran in Children (2 to Less Than 12 Years) With Homozygous Familial Hypercholesterolemia and Elevated LDL-cholesterol

12. Clinical Framework (HoFH Specific)

Primary Condition: Homozygous Familial Hypercholesterolemia (HoFH) **Diagnosis Threshold:** Fasting LDL-C > 130 mg/dL (3.4 mmol/L) with genetic confirmation **Medical Methodology:** Optimal stable dose of statin with/without other standard of care lipid-lowering therapy (LLT) **Optimization Target:** LDL-C targeted percentage reduction

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for Pediatric HoFH **Education Protocol (HCP):** Protocol training, pediatric dosing criteria, and apheresis schedule maintenance **Education Protocol (Patient/Caregiver):** Informed consent/assent procedures, injection site care, and dietary/medication adherence tracking **Quality Audit Mechanism:** Central lab sample integrity tracking & caregiver therapy adherence monitoring

14. Observation & Follow-Up Schedule

Total Duration: 720 Days (approx. 104 Follow-up weeks) **Onsite Visit Cadence:** Baseline, Day 90, Day 330, and up to Day 720 endpoints **Remote Monitoring Frequency:** Intermittent safety follow-ups as defined by site SOPs **Checklist Review Interval:**

Regular review of concurrent medications and stable apheresis status

15. Sign-off & Approval